# The Periodogram

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# Key Concepts

1) The periodogram estimate of the power spectrum is based on the magnitude squared of the discrete Fourier transform coefficients.

2) The mean of the periodogram asymptotes to the true power spectrum as the data length increases to infinity.

3) The variance of the periodogram approaches the square of the true power spectrum as the data length increases to infinity.

4) Typically the periodogram is computed as a density.

a) The amplitude is normalized so that the area under the periodogram is equal to the total power or variance of the process.

b) Different frequency units require different normalization.

# Video Screenshots

# Power Spectrum

$r_{xx}[k] \xleftrightarrow{\text{DTFT}} S_{xx}(\omega)$



Approximate $r_{xx}[k] \approx \frac{1}{L}\sum_{n=0}^{L-1} x[n]\,x[n-k]$

$$\hat{S}_{xx}(\omega) = \frac{1}{L}\sum_{k}\sum_{n} x[n]\,x[n-k]\,e^{-jk\omega}$$

let $\ell = n-k$

$$= \frac{1}{L}\sum_{\ell}\sum_{n} x[n]\,x[\ell]\,e^{-jn\omega}\,e^{j\ell\omega}$$

$$= \frac{1}{L}\underbrace{\left(\sum_{n} x[n]e^{-jn\omega}\right)}_{X(e^{j\omega})}\underbrace{\left(\sum_{\ell} x[\ell]e^{-j\ell(-\omega)}\right)}_{X(e^{-j\omega}) = X^{*}(e^{j\omega})}$$

$$= \frac{1}{L}\,|X(e^{j\omega})|^{2}$$

## General Approach

1) Window $x[n]$ : $x_w[n] = w[n]\,x[n]$, $0 \le n \le L-1$

2) Take DFT : $x_w[n] \overset{DFT}{\longleftrightarrow} X_w[k]$

3) Power Spectrum estimate: $\hat{S}_{xx}[k] = \frac{1}{LF} |X_w[k]|^2$

Periodogram: $w[n]$ given by rectangular window

Modified Periodogram: $w[n]$ any other window

Normalization

$$F = \frac{1}{L} \sum_{n=0}^{L-1} (w[n])^2$$

# Properties of the periodogram



4:35

$$\hat{S}_{xx}^{per}(\omega) = \frac{1}{LF}|X_w(e^{j\omega})|^2 \; ; \qquad x[n]w[n] \overset{DTFT}{\longleftrightarrow} X_w(e^{j\omega})$$

## 1) Mean

$$E\{\hat{S}_{xx}^{per}(\omega)\} = \frac{1}{2\pi LF}\int_{-\pi}^{\pi} S_{xx}(\theta)\, C_{ww}(e^{j(\omega-\theta)})\, d\theta$$

where $C_{ww}(e^{j\omega}) = |W(e^{j\omega})|^2$

$$w[n] \overset{DTFT}{\longleftrightarrow} W(e^{j\omega})$$

As window length L increases -

- $C_{ww}(e^{j\omega})$ is increasingly concentrated about $\omega = 0$

and

$$E\{\hat{S}_{xx}^{per}(\omega)\} \to S_{xx}(\omega)$$

Periodogram is asymptotically unbiased!

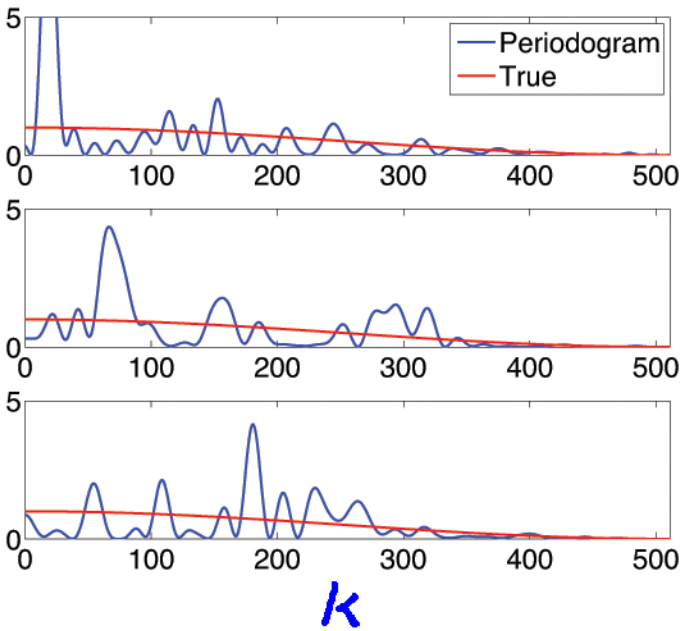
## 2) Variance

$$\mathrm{Var}\{\hat{S}_{xx}^{per}(\omega)\} \longrightarrow S_{xx}^2(\omega)$$

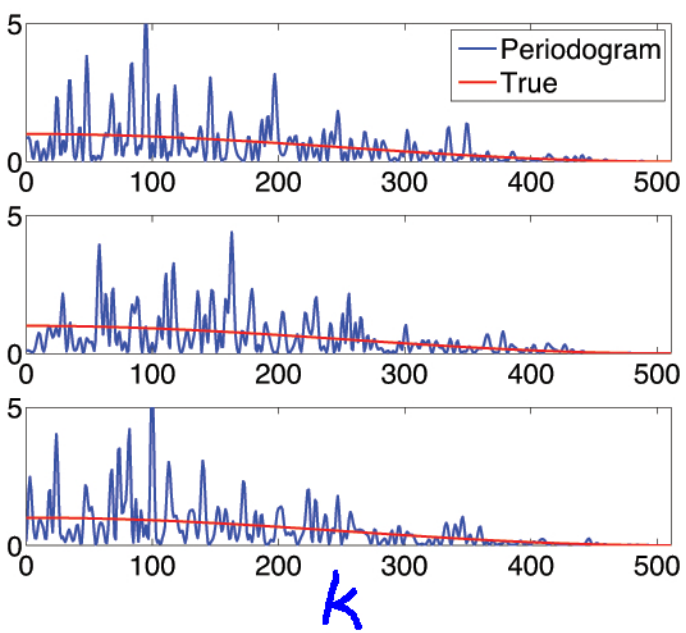
Periodogram is not a consistent estimator

Example: Three trials for different $L$ : $\hat{S}_{xx}[k] = \frac{1}{L}|X[k]|^2$, $k = 0, \ldots, 1023$

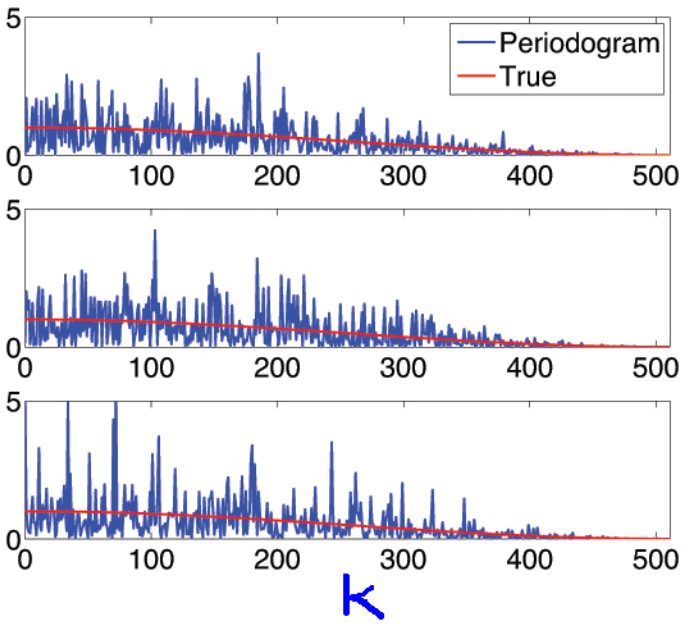
$L = 64$




$L = 256$




$L = 1024$



# Units -

- Normally (e.g. MATLAB "periodogram") displayed as a density

$$\frac{\text{Watts (or dB)}}{\text{frequency units}}: \quad \hat{S}_{xx}(\omega) \ \frac{\text{Watts}}{\text{Rads/sample}}, \quad \hat{S}_{xx}(f) \ \frac{\text{Watts}}{\text{Hz}}$$

- Amplitude is normalized so area under power spectrum is equal to total power

$\hat{S}_{xx}(\omega): 0 \le \omega < 2\pi$

$$\hat{S}_{xx}(\omega) = \frac{1}{2\pi L}\left|\sum_{n=0}^{L-1} x[n] e^{-j\omega n}\right|^2$$

$$\hat{S}_{xx}\left[\frac{k2\pi}{N}\right] = \frac{1}{2\pi L}|X[k]|^2$$

$k = 0, 1, \ldots, N-1$

$\hat{S}_{xx}(\omega): 0 \le \omega < \pi$

$$\hat{S}_{xx}(\omega) = \frac{1}{\pi L}\left|\sum_{n=0}^{L-1} x[n] e^{-j\omega n}\right|^2$$

$$\hat{S}_{xx}\left[k\frac{2\pi}{N}\right] = \frac{1}{\pi L}|X[k]|^2$$

$k = 0, 1, \ldots \frac{N}{2} - 1$

$\hat{S}_{xx}(f): 0 \le f < \frac{f_s}{2}$

$$\hat{S}_{xx}(f) = \frac{2}{f_s L}\left|\sum_{n=0}^{L-1} x[n] e^{-j\frac{2\pi f n}{f_s}}\right|^2$$

$$\hat{S}_{xx}\left[\frac{kf_s}{N}\right] = \frac{2}{f_s L}|X[k]|^2$$

$k = 0, 1, \ldots \frac{N}{2} - 1$

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